Rickson Raphel - AI Engineer

ricksonraphelofficial@gmail.com | +91 9048614684 | LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI Engineer with 1.5 years of experience building and deploying production LLM applications, including RAG pipelines, LangGraph agent workflows, real-time voice agents and multimodal AI. Skilled in Python, FastAPI, hybrid retrieval, LLM evaluation and OpenAI and Gemini models.

PROFESSIONAL EXPERIENCE

AI Engineer — **Oliveboard** (exam-prep platform), Bengaluru May 2025 – Present
Employee of the Quarter — Q1 2026

- **Vidya** — Built an automated document generation pipeline (FastAPI + OpenAI Batch API) with a LangGraph agent, reducing generation time by **~80%** across multiple content domains.
- **TranslateLab v2** — Engineered a multilingual document processing pipeline supporting **12 languages** with a human review step before translation, reducing end-to-end turnaround time by **~85%**.
- **Hybrid RAG Pipeline** — Built a ChromaDB + BM25 + FlashRank retrieval system grounded on enterprise knowledge bases, improving answer accuracy by **85%**.
- **Evaluation Harness (mcq-eval-test):** Built a generate, check, retry and escalate pipeline that validates generated questions and informs difficulty calibration in Vidya.
- **PYP Automation Suite** — Automated document parsing and structured data extraction via GPT-4.1, reducing processing time by **~98%**.
- **LPT Evaluator** — Built a multimodal AI evaluation system using Gemini Vision and Audio to assess reading, writing and comprehension responses automatically, eliminating manual scoring overhead. Also built a batch transcription service with per-file offset merging for large audio.
- **DevOps & Deployment** — Deployed and maintained 9+ internal tools as production-grade services on Ubuntu with Nginx and Gunicorn, with Systemd auto-healing and SSL/TLS for secure WSS.

TECHNICAL SKILLS

Languages: Python, SQL, Java
AI & GenAI: LLMs, RAG Architectures, Agentic Workflows, Graph-based Agents, LLM Evaluation, Eval Harnesses, Feedback Loops, ChromaDB, BM25, LangChain, LangGraph, API Integration, Prompt Engineering, TTS Integration
Backend & Systems: FastAPI, Flask, WebSockets, Redis Pub/Sub, asyncio, Nginx, Gunicorn, Systemd
Data & Storage: PostgreSQL, SQLite, asyncpg, PyMuPDF
Tools & Platforms: Git, Docker, Linux (Ubuntu), Postman, VS Code

PROJECTS

- **Voice Mentor** — Built a full-duplex real-time voice agent using OpenAI Realtime API and Cartesia TTS, with a low-latency WebSocket pipeline for raw Float32 PCM audio, grounded by hybrid RAG.
- **Yantra** — Local personal AI assistant built on Ollama LLMs, ChromaDB RAG and PostgreSQL, deployed as a PWA via ngrok with an MCP-based system control terminal.

EDUCATION

B.Tech in AI & Data Science — Viswajyothi College of Engineering and Technology CGPA: 8.11